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Palaeomagnetic, geochronological and geological constraints on the tectonic evolution of the Mejillones Peninsula, northern Chile

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Palaeomagnetic and geochronological data from an Early Palaeozoic high grade metamorphic complex (Jorgina Formation) and Jurassic layered basic intrusion (Moreno Complex) are reported from the Mejillones Peninsula of northern Chile (23–23°30′S). ^{40}Ar–^{39}Ar dates from the Lower Palaeozoic Jorgina Formation and the Moreno Complex are between 170 and 158 Ma, coincident with a phase of emplacement of the north Chilean coastal batholith. This suggests that intrusion and magnetization of the Moreno Complex and the metamorphism and remagnetization of the Jorgina Formation were related to batholith emplacement. Extracted stable components of magnetization from all units (17 sites) define site-mean directions with a scattered distribution. The scatter in site-mean directions is interpreted as being due to minor, localized, non-uniform, block-fault related (normal or strike-slip, or both) rotation after 158 Ma. The palaeomagnetic and geochronological data indicate that no significant large-scale latitudinal translation of crustal blocks has taken place in this part of northern Chile since the Late Jurassic. In addition, they indicate that the uniform clockwise rotation after the mid-Cretaceous which affected the adjacent Cordillera de la Costa either did not extend into the Mejillones Peninsula or took the form of localized block-fault rotations. The restriction of palaeomagnetically defined styles of rotation to discrete areas within the north Chilean forearc indicates that forearc wide block-fault rotation models are not applicable to the Pacific margin of northern Chile.

KEY WORDS Northern Chile Mejillones Peninsula Palaeomagnetism Cordillera de la Costa Andes Block rotation

1. INTRODUCTION

The present day configuration of the Andean orogenic belt is largely a product of the well known Late Triassic–Recent Andean orogenic cycle (e.g. Coira *et al.* 1982). Differences in the style of orogenesis during the Andean cycle have led to marked internal heterogeneities within the mountain chain (Gansser 1973; Figure 1). In particular, the northern Andes (north of 3°S) of Colombia and Ecuador differ considerably from the central Andes (3–45°S) of Peru and Chile. The northern Andes are underlain by oceanic crust (Case *et al.* 1971; Lonsdale 1978) and are thought to have formed through the accretion of discrete fragments of oceanic crust during the Mesozoic and Cenozoic (McCourt *et al.* 1984; McGeary and Ben-Avraham 1985; Aspden and McCourt 1986; Feininger 1986; Roperch *et al.* 1987). In contrast, the central Andes are underlain by continental crust and are considered to represent a marginal orogen formed through subduction since Late Triassic times (James 1971; Coira *et al.* 1982). Palaeomagnetic studies indicate that Jurassic and Cretaceous rocks of the Andean margin have not undergone any significant latitudinal

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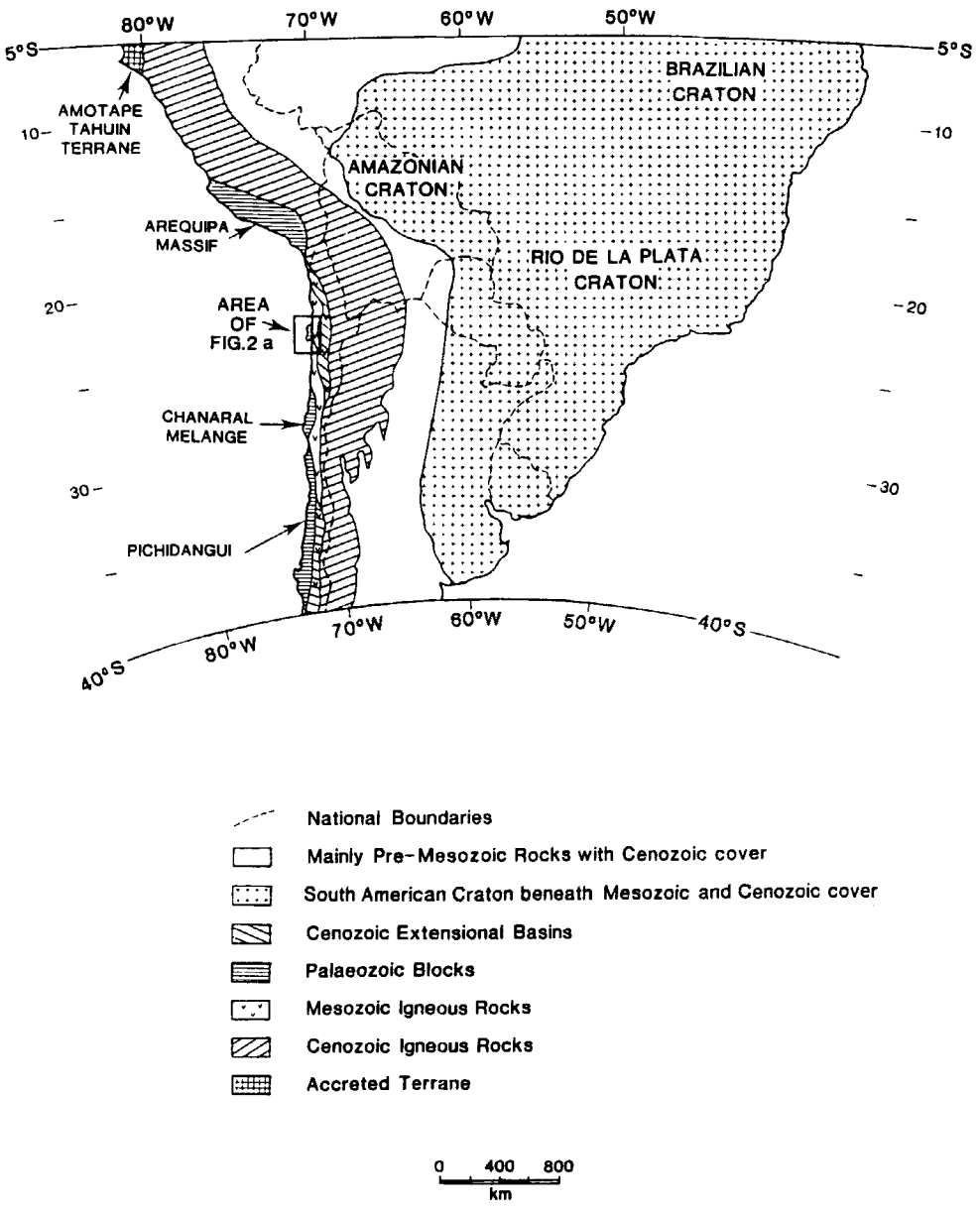


Figure 1. Simplified map of the lithotectonic units of central South America

displacement, but have either suffered 'localized' *in situ* block rotation (see Beck 1988 for a review) or are concordant with the apparent polar wander path (APWP) for the Mesozoic of South America (e.g. Irwin *et al.* 1987). These results have led to a general belief that accretion tectonics has had no role in Andean orogenesis (e.g. Dalziel and Forsythe 1985), a hypothesis supported by faunal studies (Hallam 1986). However, the presence of a number of potential Palaeozoic suspect terranes (Howell *et al.* 1985; Ramos *et al.* 1986; Ramos 1988) points to terrane accretion having a significant influence on pre-Andean orogenesis. The identification of an accreted continental terrane in north-west Peru (Mourier *et al.* 1988) suggests that terrane accretion may have been significant in central Andean Mesozoic orogenesis.

This paper describes the palaeomagnetism, geochronology and geology of a basic igneous and metamorphosed igneous massif from the north Chilean Pacific margin (Mejillones Peninsula; Figure 2). The

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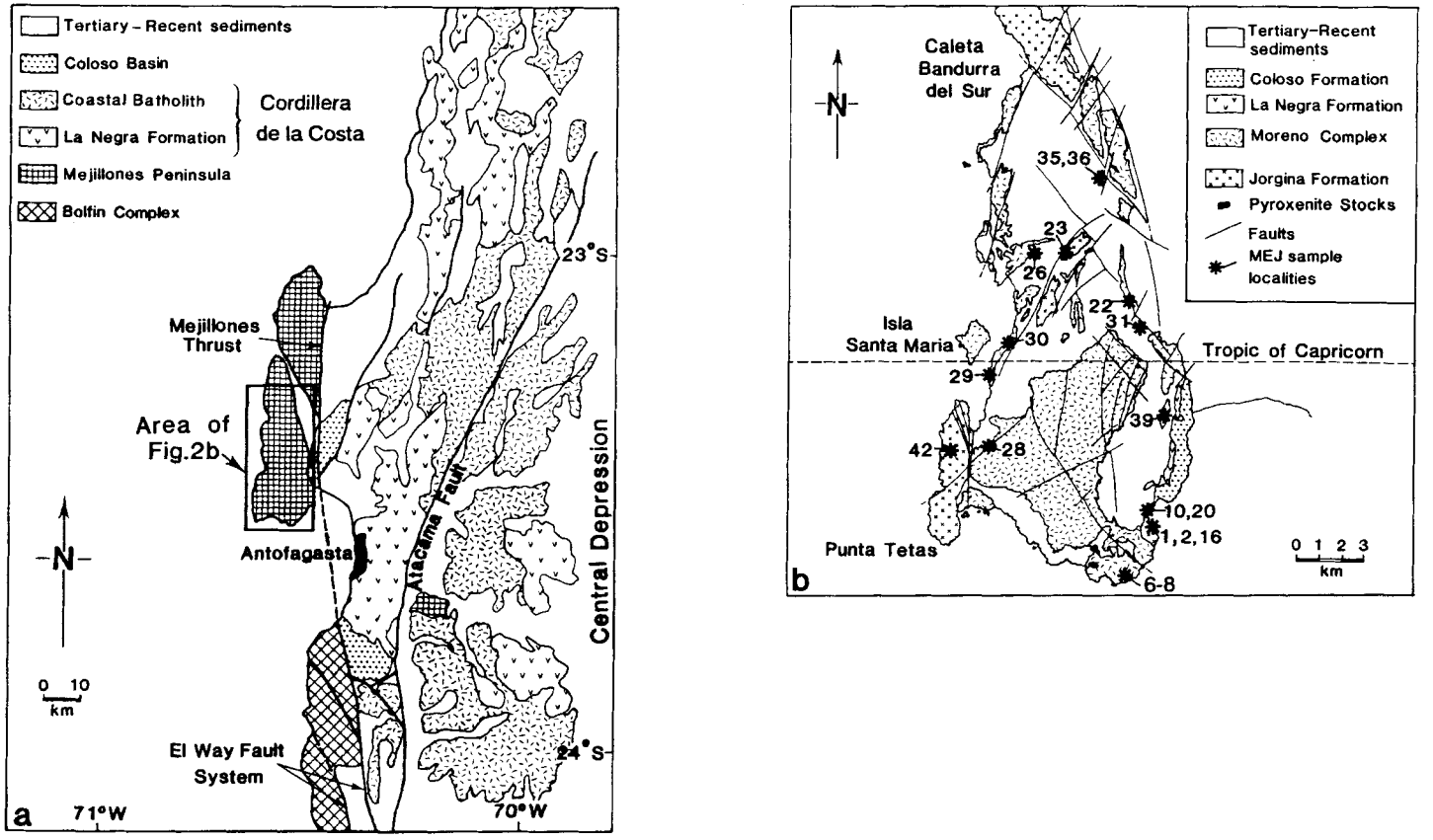


Figure 2. Geological map of the study area showing the aerial extent of the Coloso Basin. (b) Geological map of the southern part of the Mejillones Peninsula showing palaeomagnetic and radiometric sample localities

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massif consists of a discrete geological unit (Ferraris and Di Biase 1978; Damm *et al.* 1986), which lies outboard of the Mesozoic acidic volcanics and plutonics of the adjacent Cordillera de la Costa (Figure 2). The evidence indicates that the massif has undergone localized, *in situ*, non-uniform block rotation since the Late Jurassic, in contrast to the uniform clockwise rotation which affected the adjacent Cordillera de la Costa after the mid-Cretaceous (Turner *et al.* 1984a; Hartley *et al.* 1988).

2. GEOLOGICAL BACKGROUND

2a. Mejillones Peninsula

The Mejillones Peninsula is separated from the Chilean mainland by a NNE–SSW oriented shear zone (Figure 2) which forms a strand of the major Atacama Fault Zone (Figure 2a). The shear zone has been interpreted as a major Mesozoic thrust (the Mejillones Thrust; Hartley *et al.* 1988), although Late Mesozoic sinistral movement along the shear zone was also significant (Scheuber and Andriessen 1990) and vertical movements currently dominate (Okada 1971; Armijo and Thiele 1990). The peninsula is composed of igneous and metamorphosed igneous (and ?sedimentary) rocks, exposed in two fault-bounded blocks trending NW–SE. The northern part consists of garnet–quartz–mica schists, migmatitic gneisses, acidic intrusives, quartzites and subordinate amphibolites of the Jorgina Formation (Buchelt and Zeil 1986; Baeza and Pichowiak 1988), which were subject to contact metamorphism related to the intrusion of an Early Jurassic tonalitic granodiorite (Baeza and Venegas 1985; Diaz *et al.* 1985). A 530 Ma Rb–Sr whole rock isochron was obtained from the mica schists (Diaz *et al.* 1985) and interpreted as representing an original age of crystallization. A U–Pb zircon date of 561 + 12/−14 Ma (defined from the upper intercept of the concordia diagram) was obtained from a granodioritic–tonalitic intrusion in the north of the Peninsula (Damm *et al.* 1986).

The southern part of the peninsula is dominated by an Early Jurassic layered intrusion — the Moreno Complex (Damm *et al.* 1986; Figure 2). It consists of layered quartz–hornblende gabbros and pyroxenite–norite stocks, which are similar geochemically to the basaltic andesites of the La Negra Formation which form the bulk of the Cordillera de la Costa (Damm *et al.* 1986; Baeza and Pichowiak 1988; Figure 2). The Moreno Complex was intruded into gneisses and amphibolites of the Jorgina Formation (Figure 2). A Sm–Nd whole rock isochron for amphibolites of the Jorgina Formation gave a date of 521 ± 55 Ma (Damm *et al.* 1986), which compares favourably with the Cambrian dates obtained for the formation from the northern part of the peninsula and derived from different dating techniques.

In the south-east of the peninsula the Moreno Complex is unconformably overlain by poorly exposed basaltic andesites of the mid-Jurassic La Negra Formation and conglomerates of the Lower Cretaceous Coloso Formation (Ferraris and Di Biase 1978; Baeza and Venegas 1984). Bedding, where visible, is generally subhorizontal although locally increases up to 50° in dip occur close to inferred faults.

Two dominant fault trends (N/NE–S/SW and N/NW–S/SE) occur throughout the peninsula (Figure 2). However, many of these faults have been mapped from aerial photographs (e.g. Buchelt and Zeil, 1986) and are particularly poorly exposed within the peninsula due to weathering. Where seen in coastal exposures, the faults cut Pleistocene marine sediments and have a normal geometry (Armijo and Thiele 1990) with dips of 60–80° to the E/ENE or W/WSW. Marine terraces of Late Cenozoic age occur at heights of 600 m on the fault blocks (Okada 1971), indicating that substantial Late Cenozoic vertical uplift has affected the peninsula.

2b. Cordillera de la Costa

The Cordillera de la Costa forms a prominent topographic high throughout northern Chile. It is composed of the 10 km thick andesite-dominated La Negra Formation (Garcia 1967) and associated granitic intrusions. Rogers (1985) obtained a single K/Ar whole rock mid-Jurassic date of 186 ± 14 Ma from the formation. To the east, the La Negra Formation unconformably overlies Triassic and Palaeozoic sediments and is

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interbedded with shallow marine clastics and carbonates of Lower to Mid-Jurassic age (Ferraris and Di Biase 1978).

Jurassic granodioritic and granitic plutons ranging in age between 200 and 135 Ma (Ferraris and Di Biase 1978; Halpern 1978; Berg and Breitkreuz 1983; Rogers 1985; Damm *et al.* 1986) intrude the La Negra Formation. The plutons are thought to represent a granitic batholith emplaced within the Cordillera de la Costa (Ulriksen 1979). The La Negra Formation and associated granodiorites have been interpreted as the remnants of a magmatic arc which extended from southern Peru to central Chile (James 1971; Coira *et al.* 1982). Conversely, Rogers (1985) considers that the La Negra Formation was erupted through a series of fissures within an ensialic backarc basin.

In Antofagasta Province, the La Negra Formation and Jurassic plutons are overlain unconformably by sediments of the Lower Cretaceous Coloso Basin (Ferraris and Di Biase 1978; Hartley 1987). The basin consists of a 3 km thick continental conglomerate sequence, dominated by proximal to distal alluvial fan facies sourced from the west (Flint *et al.* 1986; Flint and Turner 1988). These pass upwards conformably into the shallow marine sandstones and limestones of the El Way Formation, which contain Hauterivian macrofossils (Jurgan 1974). Along the coastal margin, Pliocene–Recent aeolian and shallow marine sandstones overlie all older rocks unconformably.

3. GEOCHRONOLOGY

3a. Experimental details

Five hornblende separates were analysed by the $^{40}Ar/^{39}Ar$ method. Analyses were performed at the geochronology laboratory of the University of Leeds. Full details of the irradiation and argon analysis used at Leeds are given by Parsons *et al.* (1988). The samples were purified using a Franz magnetic separator and the final 100% purity of the *ca.* 75 mg samples were checked under a binocular microscope. The samples were irradiated at AWRE (Aldermaston, UK) in the Herald reactor. The international standard minerals, HB3GR hornblende and LP6 biotite, together with the Leeds laboratory standard hornblende FY12a were used as *J* valve monitors for the irradiation (Roddick 1983). Errors quoted on the original heating steps are analytical errors whereas errors quoted on the total fusion ages include the *J* value errors (Table 1).

Table 1. Geochronological data from the Mejillones Peninsula

Unit	Sample No.	Gas forming (%)	Plateau age (Ma)	Integrated age (Ma)
Jorgina Formation	MEJ23	81·9	170 ± 4	172 ± 4
Jorgina Formation	MEJ44	53·0	162 ± 3	170 ± 3
Moreno Complex	MEJ6	71·8	158 ± 6	189 ± 6
Moreno Complex	MEJ32	94·0	167 ± 4	164 ± 4
Moreno Complex	MEJ38	83·9	161 ± 6	188 ± 6

3b. $^{40}Ar/^{39}Ar$ data

$^{40}Ar/^{39}Ar$ data obtained from pure hornblende separates from the Jorgina Formation and Moreno Complex are presented in Table 7. Spectra from the Moreno complex are both U-shaped (MEJ6, MEJ38; Figure 3), indicating the presence of excess argon (Lanphere and Dalrymple 1976), and step-shaped (MEJ32), indicating that the hornblende had undergone some argon loss due to reheating. Spectra from the Jorgina Formation are either step-shaped (MEJ44), indicating reheating, or have a well defined plateau (MEJ23).

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All the spectra have plateau ages between 158 ± 6 and 170 ± 4 Ma which reflects the time at which the hornblendes cooled below their closure temperatures.

The results indicate that the Moreno Complex cooled below the closure temperature of hornblende (450–550°C; Harrison 1981) between 167 and 158 Ma. The reheating of the originally Cambrian Jorgina Formation took place at roughly the same time (between 170 and 162 Ma). These dates correspond well with the main phase of emplacement of the north Chilean batholith (175–150 Ma), suggesting that the intrusion of the Moreno Complex and reheating of the Jorgina Formation were associated with batholith emplacement.

4. PALAEOMAGNETIC ANALYSIS

4a. Sampling and laboratory methods

Oriented hand samples and drill cores from 17 sites, yielding 87 specimen cores, were collected from the Moreno Complex and Jorgina Formation in the southern part of the Mejillones Peninsula (Figure 2). Where possible, care was taken to avoid sampling close to faults. Specimens were treated by stepwise thermal demagnetization and measured using a two-axis cryogenic magnetometer [see Hartley *et al.* (1988) for further details]. Isolated components of magnetization were extracted using principal component analysis (Kirschvink 1980) to calculate the least-squares best fitting line with an angular deviation of less than 10°.

4b. Results

Analysis of thermal demagnetization data from the Moreno Complex (Table 2) showed that specimen behaviour could be divided into two main categories: (1) specimens which showed essentially single component magnetizations (85%) and (2) those with two or more components of magnetization (15%).

Examples of category (1) are shown in Figure 4, where the normalized intensity decay curve for MEJ13·1 (gabbro) decreases steadily during thermal demagnetization to 580°C, the Curie temperature of magnetite, where there is an increased loss of natural remanent magnetization (NRM). The orthogonal vector plot indicates a single component of magnetization with $D = 330$, $I = -22$. Other specimens which show single components of magnetization have a wide range of declinations.

Category (2) magnetizations are illustrated in Figures 5 and 6. Figure 5 shows the thermal demagnetization results of MEJ7.1A. The initial NRM direction lies in the north-east quadrant and above 400°C moves to the north-west quadrant (Figure 5a). Note that after an initial increase in normalized intensity there are marked unblocking temperatures at 400 and 500°C (Figure 5b). The orthogonal vector plot shows two separate components with $D = 30$, $I = -18$ and $D = 220$, $I = +15$. Similar results are shown for MEJ36.2 (Figure 6). From an initial steep, upward inclination, the NRM directions move towards the south-west and the NRM intensity shows a slight increase rather than loss during this movement (Figure 6b). On the orthogonal plot two discrete components are discernible; their respective positions are $D = 45$, $I = -22$ and $D = 238$, $I = -30$.

Analysis of thermal demagnetization data from the Jorgina Formation revealed very similar behaviour to that of the Moreno Complex, with the presence of one or two components of magnetization in individual specimens. Figure 7 illustrates MEJ20.1, a gneiss from the Jorgina Formation, with a single component of magnetization. The normalized intensity decay curve decreases steadily up to 580°C, where there us an increased loss of NRM. The orthogonal vector plot indicates a single component of magnetization with $D = 10$, $I = -5$. Note that this is shallower and eastwards of the present geomagnetic direction for the area.

Extracted components of magnetization for the Moreno Complex and Jorgina Formation are shown in Figure 8a and b for specimen and site means, respectively. Note that the directions show a similar

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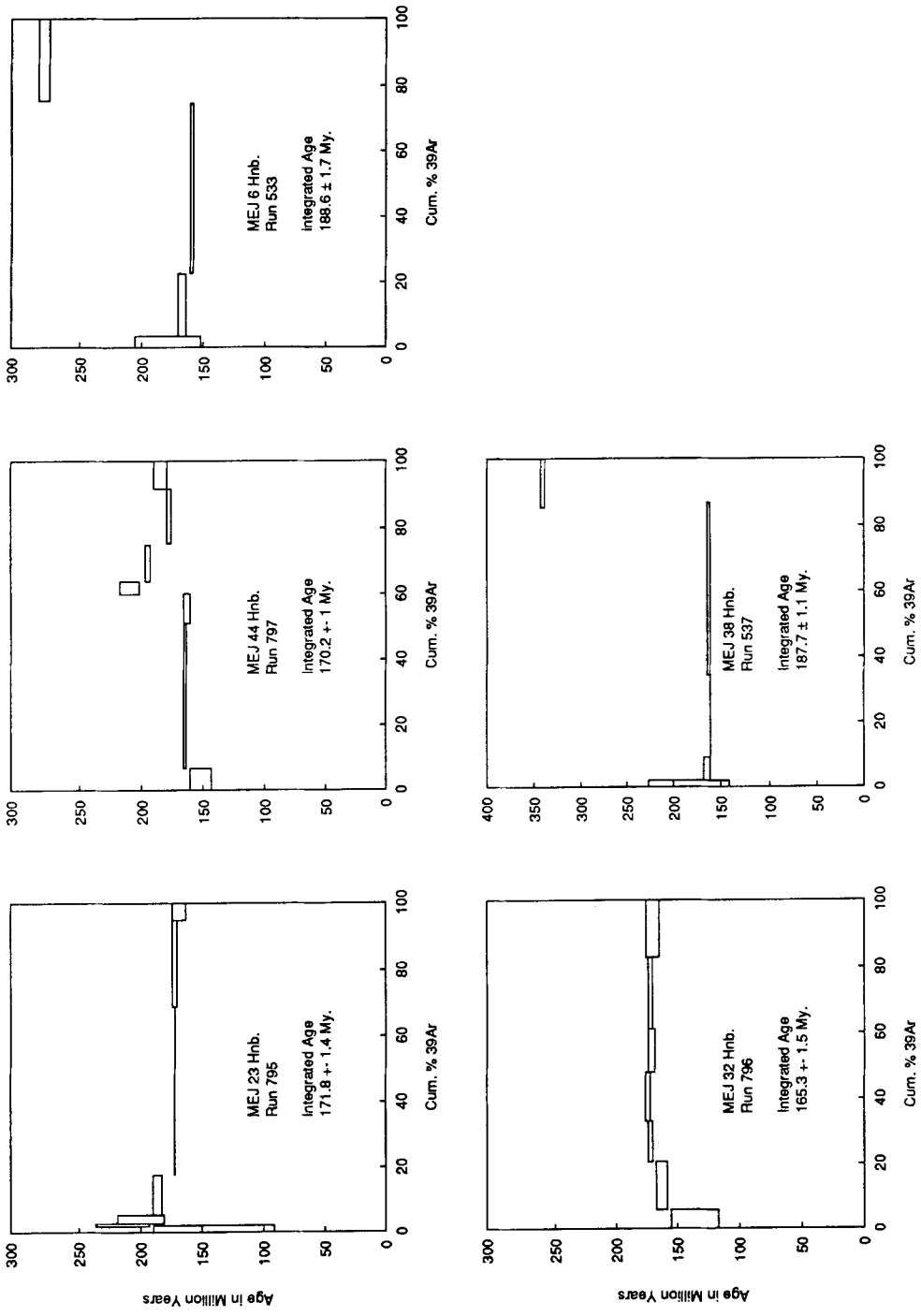


Figure 3. Heating spectra for MEJ23, MEJ44 (Jorgina Formation), MEJ6, MEJ32 and MEJ38 (Moreno Complex) showing cumulative percentage of ^{39}Ar released per heating step and the resultant integrated age (see text)

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Table 2. Components of magnetization extracted from the Mejillones Peninsula. Statistics after Fisher (1953).

Site	n	R	k	a_{95}	Dec (°)	Inc (°)	Range (°C)
Jorgina Formation:							
MEJ1	4	3·897	29·0	17·2	328	−36	0–500
MEJ2	5	4·831	23·7	16·0	348	−45	0–500
MEJ10	5	4·981	222·0	5·2	351	−42	0–500
MEJ16	5	4·980	204·0	5·4	261	+13	0–500
MEJ20	5	4·987	319·2	4·3	11	−25	0–550
MEJ23	5	4·798	19·9	17·0	46	−11	0–550
MEJ42	5	4·971	138·1	6·6	189	−36	0–500
Moreno Complex:							
MEJ7	5	4·979	196·4	5·5	34	−29	0–400
MEJ8	7	6·578	14·2	16·6	308	−58	150–500
MEJ22	4	3·963	81·0	10·2	314	−40	0–550
MEJ28	6	5·473	9·5	22·9	276	−47	150–500
MEJ29	5	4·891	36·8	12·8	5	−9	150–500
MEJ30	4	3·881	25·1	18·6	345	−27	0–300
MEJ31	6	5·983	302·1	3·9	23	+3	0–500
MEJ35	4	3·997	878·8	3·1	263	+44	100–580
MEJ36	6	5·622	13·2	19·1	251	−32	150–550
MEJ39	6	5·433	8·8	23·9	40	−46	150–400

N = number of specimens prepared from orientated block samples;
R = length of resultant vector;
k = best estimate of Fisher precision parameter;
a_{95} = radius of cone 95% confidence about the site-mean direction;
Inc and Dec are site-mean inclination and declination, respectively;
range (°C) = temperature range over which component analysis was under taken.

trend to the normal-reversed directions derived from the Cordillera de la Costa (Hartley *et al.* 1988) and the present geomagnetic direction for the area (Figure 8b).

Isothermal remanent magnetization (IRM) acquisition curves (Figure 9), together with reflected light microscopy, were used to determine the magnetic mineralogy of samples. Isothermal remanent magnetization curves (Dunlop 1972) allow discrimination between rocks containing low coercivity minerals such as magnetite which normally saturate in applied fields of less than 0·3 T and high coercivity minerals such as haematite for which fields in excess of 5 T may be required for saturation (Turner *et al.* 1984b). Both techniques indicate that the magnetic carrier in both the Moreno Complex and the Jorgina Formation is magnetite as no increase in the IRM was observed above an applied field of 0·4 T (Figure 9). In the Moreno Complex magnetite occurs as octahedral phenocrysts within the groundmass of the samples, whereas samples from the Jorgina Formation contain magnetite as small disseminated euhedral crystals. In both units haematite is occasionally observed as an alteration product of unstable phyllosilicate minerals. However, the contribution of haematite to the measured remanent magnetization of the specimens is negligible as no indication of haematite is present in the IRM curves (Figure 9). Consequently, it is probable that where haematite grains are observed as alteration products the grains are not large enough to breach the paramagnetic threshold of haematite.

5. INTERPRETATION OF PALAEOMAGNETIC AND GEOCHRONOLOGICAL DATA

5a. $^{40}Ar/^{39}Ar$ systematics and acquisition of magnetization

The closure temperatures of hornblende (the temperature at which argon diffusion from the hornblende lattice ceases) lie in the range 450–550°C (Harrison, 1981), which is coincident with the temperature range over which fine grained magnetite can acquire a thermal remanent magnetization (TRM) (Irwin *et al.*

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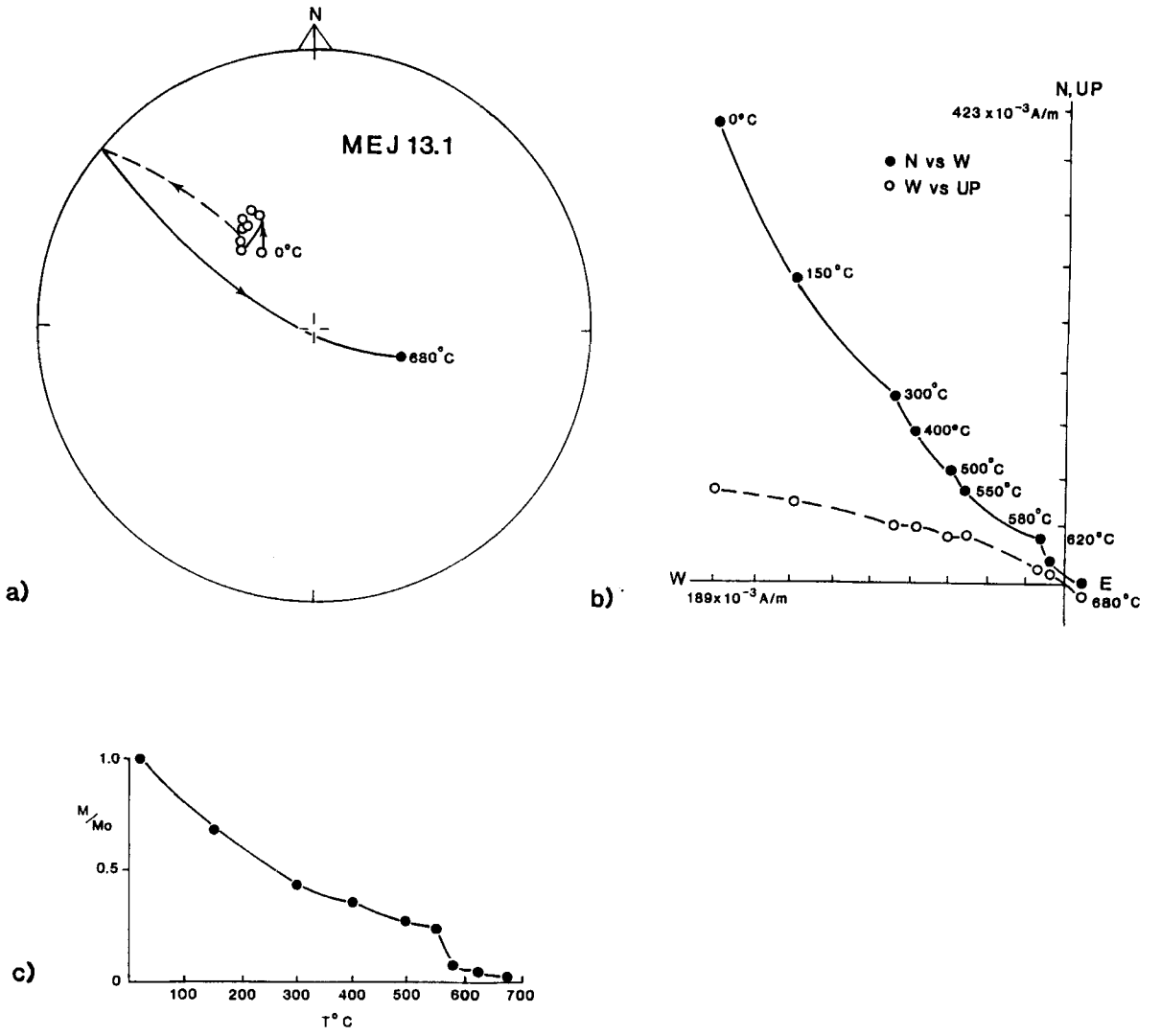


Figure 4. Palaeomagnetic results from MEJ13.1 (Moreno Complex). (a) Equal-area stereographic projection illustrating directional changes during demagnetization. Open data points are in the upper hemisphere projection; filled data points are in the lower hemisphere. (b) Orthogonal Zijderveld diagram; note different scales on vertical and horizontal axes. (c) Normalized intensity decay curve

1987). Therefore, the plateau ages of the samples defined by $^{40}Ar/^{39}Ar$ spectra (Table 1, Figure 3) are likely to record the age at which the TRM of the rocks was acquired.

The magnetization of the sampled sites is thought to be carried mainly by fine grained magnetite. The $^{40}Ar/^{39}Ar$ incremental heating age spectra indicate that the thermal history of the rocks is relatively simple with minor argon loss after the Late Jurassic (158 Ma). The relationship between hornblende closure temperatures and the acquisition of magnetization by fine grained magnetite suggests that the magnetization of the Moreno Complex and the Jorgina Formation was acquired between 170 and 158 Ma during the main emplacement phase of the north Chilean batholith. The magnetization of the Moreno Complex was acquired during cooling following intrusion, whereas the magnetization of the Jorgina Formation was acquired through cooling following metamorphism associated with batholith emplacement.

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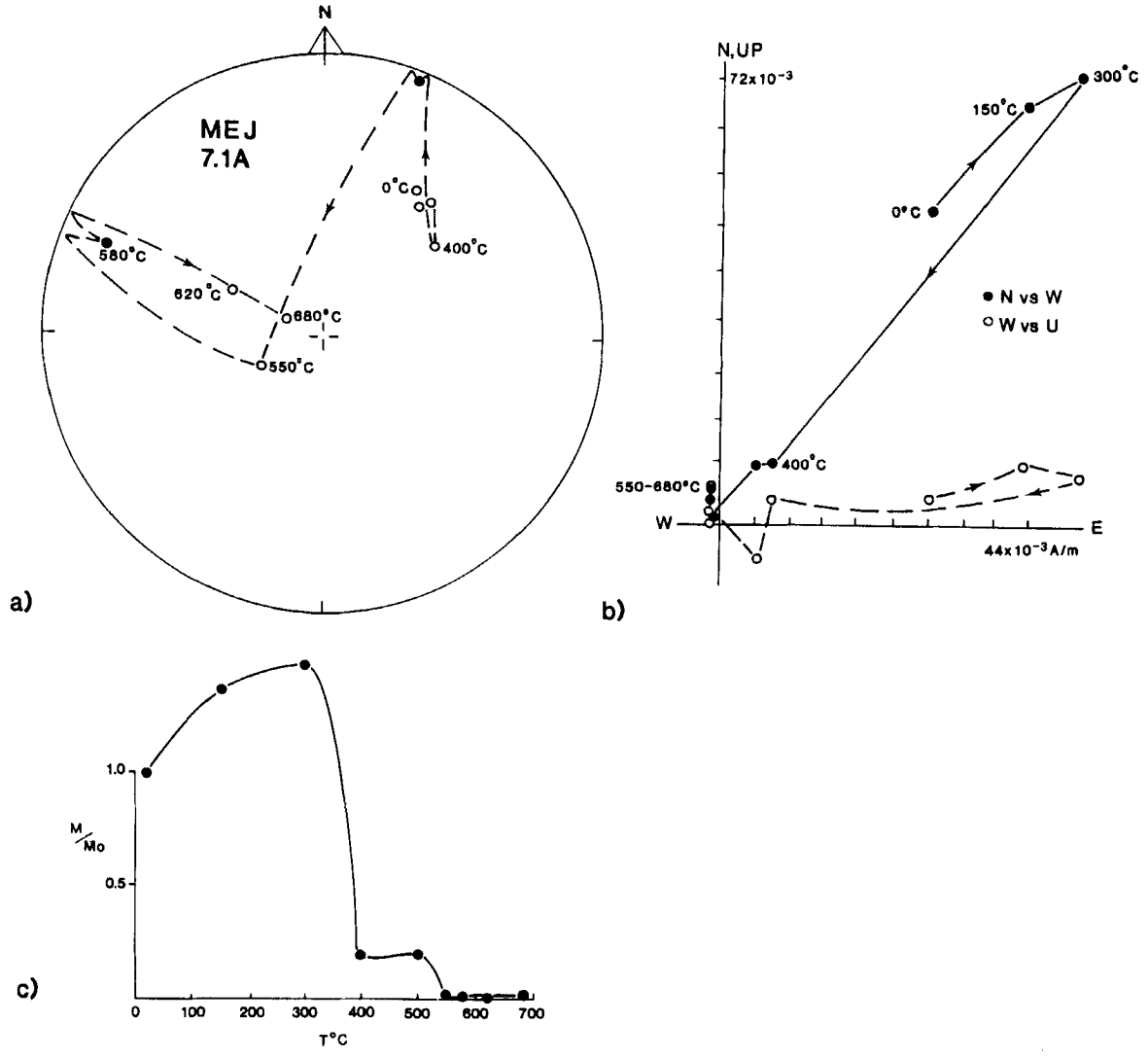


Figure 5. Palaeomagnetic results from MEJ7.1A (Moreno Complex); see Figure 4 caption for explanation

5b. Distribution of isolated components of magnetization

As previously noted, the extracted components of magnetization from the Mejillones Peninsula have a similar trend to the normal-reversed directions derived from the adjacent Cordillera de la Costa (Figure 8). The varied distribution of the isolated stable components of magnetization with a concentration close to the present geomagnetic field direction for the sampled area (Figure 8) could suggest that some of the sites have been subject to Recent remagnetization. However, this scenario is considered unlikely as the NRM of the samples is carried by magnetite, which has not been oxidized (Figure 9). Also, remagnetization of the samples would require a substantial heat source and no igneous activity has been recorded from this area of the Cordillera de la Costa after the mid-Cretaceous. It is probable, therefore, that sites with mean directions lying close to the present geomagnetic direction have suffered little or no rotation after the Late Jurassic, particularly as the Late Jurassic to present APWP for South America has not varied greatly from the present day rotation axis (Beck 1988). Sites with mean directions that are significantly displaced from the present geomagnetic direction for the area are likely to have undergone some rotation either in a clockwise (negative inclinations in the north-east quadrant or positive inclinations in the south-

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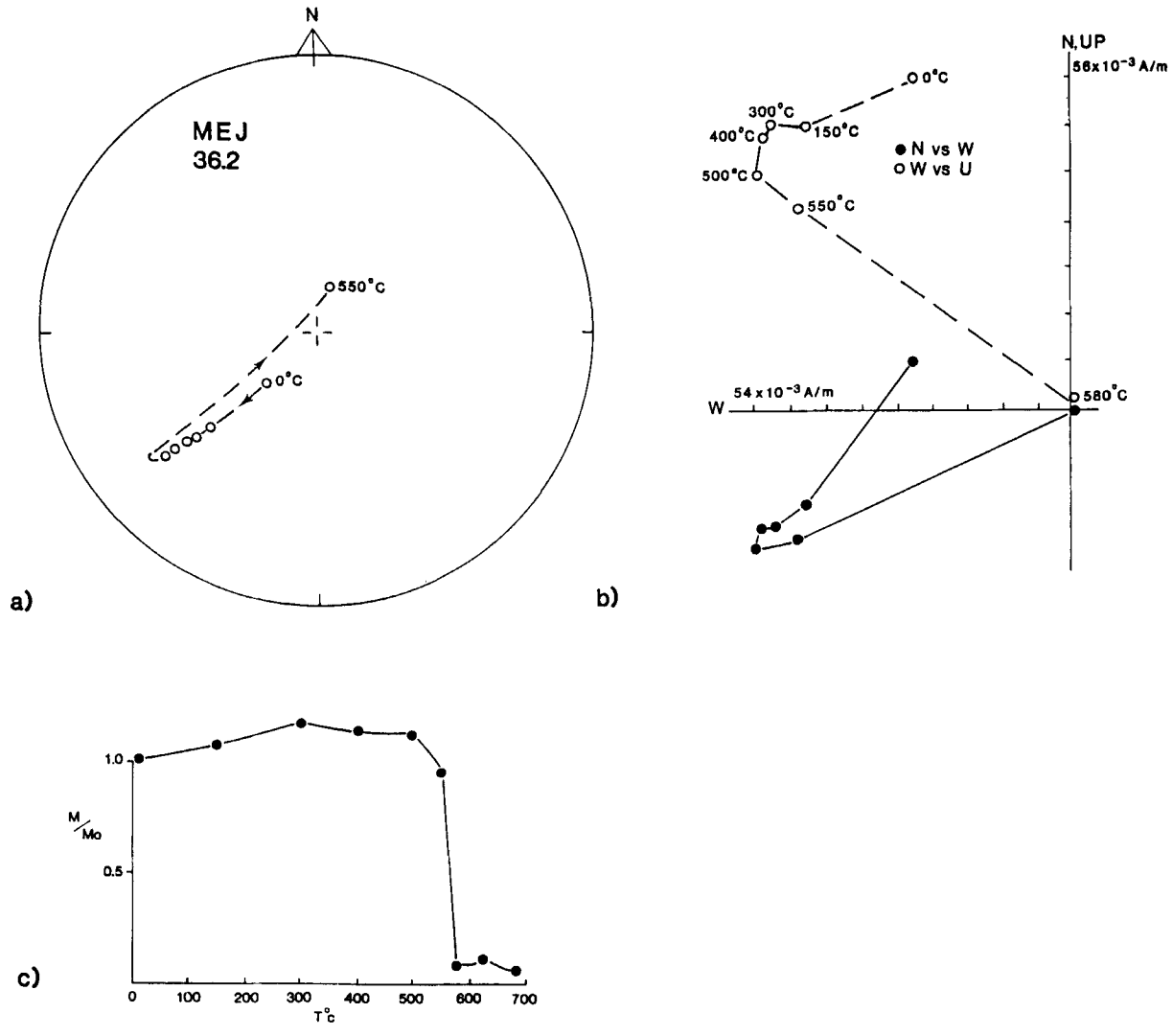


Figure 6. Palaeomagnetic results from MEJ36.2 (Moreno Complex); see Figure 4 caption for explanation

west quadrant) or anticlockwise (negative inclinations in the north-west quadrant) sense. However, the wide variation in site-mean directions indicates that this rotation is not systematic across the sampled area and is, therefore, probably related to small-scale, non-uniform, fault-block rotation. Block faulting is favoured as the most likely rotation-generating mechanism as tilting of strata deposited after the Late Jurassic is only associated with small-scale, normal faults on the Mejillones Peninsula (Ferraris and Di Biase 1978; Armijo and Thiele 1990). Unfortunately, it is impossible to constrain the age of rotation of these faults as they have had a long history of movement since the Late Jurassic and are currently active as normal faults.

The palaeomagnetic results from the Mejillones Peninsula contrast with those from the adjacent part of the Cordillera de la Costa (Hartley *et al.* 1988). The results from the Cordillera de la Costa were derived from samples of differing ages and lithologies and showed that rotation uniformly affected the sampled area and were interpreted as reflecting substantial ($24 \pm 7°$) clockwise rotation after the Early Cretaceous. The localized rotations which palaeomagnetic data suggest affected the Mejillones Peninsula also indicate that no latitudinal displacement of the peninsula has taken place since Late Jurassic times.

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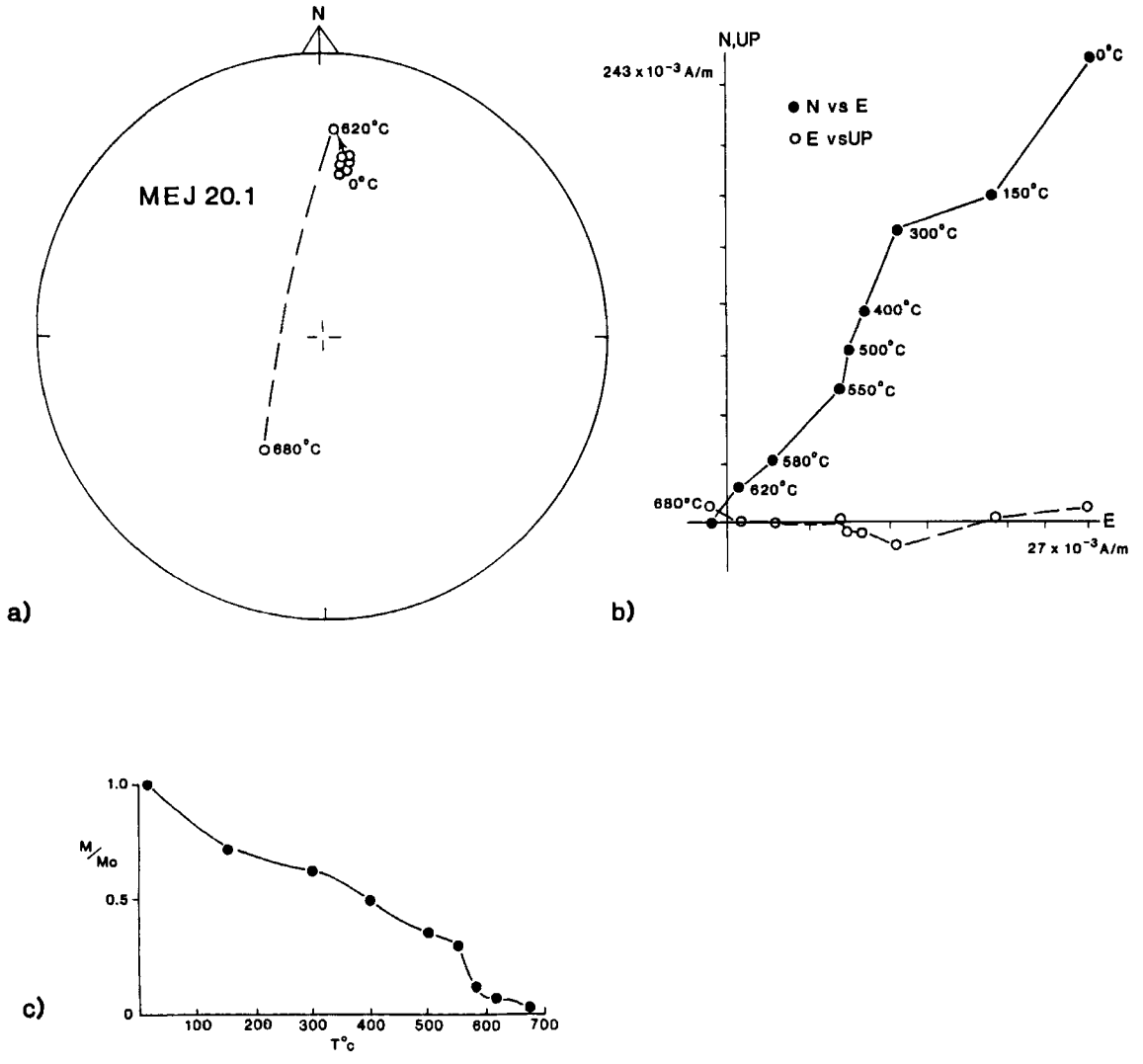


Figure 7. Palaeomagnetic results from MEJ20.1 (Jorgina Formation); see Figure 4 caption for explanation

6. DISCUSSION AND CONCLUSIONS

This study highlights some of the difficulties involved in attempting to distinguish allochthonous terranes located on the Andean coastal margin using palaeomagnetic and geochronological techniques. In particular, problems arise due to Andean orogenic events, notably the emplacement of the mid-Jurassic batholith within the Cordillera de la Costa which may have resulted in substantial remagnetization and resetting of the radiometric ages of pre-Jurassic rocks (Forsythe *et al.* 1987).

Palaeomagnetic and geochronological data from the Mejillones Peninsula suggest that no significant latitudinal displacement of the area has taken place since emplacement of the mid-Jurassic north Chilean batholith. This indicates that if terrane accretion was an important mechanism in Andean orogenesis it must have taken place before mid-Jurassic times. In addition, the non-uniform nature of rotation within the peninsula compared with the uniform rotation of the adjacent Cordillera de la Costa suggests that forearc-wide rotation models (*cf.* Beck 1988) are not applicable to northern Chile. This interpretation is supported by the presence of Early to mid-Tertiary thrust sheet related rotations in the North Chilean Precordillera 250 km to the east (Hartley *et al.* in press).

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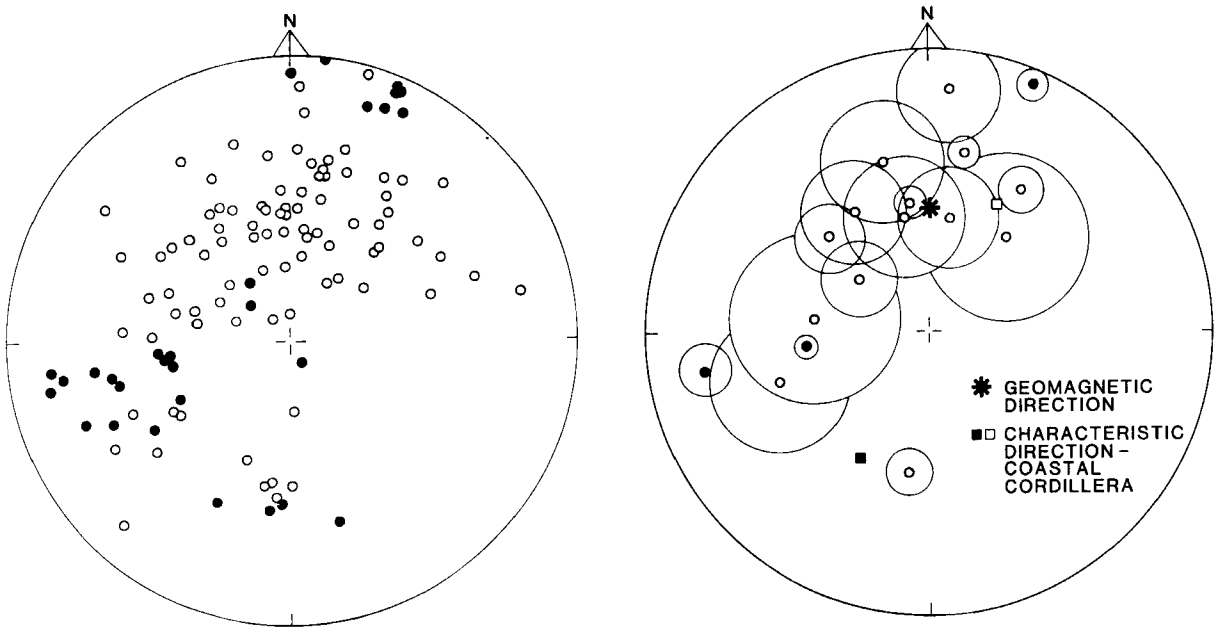


Figure 8(a). Equal-area stereographic projection of extracted components of magnetization from the Mejillones Peninsula. (b) Equal-area stereographic projection of site-mean directions and α_{95} confidence limits for the data from the Mejillones Peninsula given in Table 2. Open data points are in the upper hemisphere projection; filled data points are in the lower hemisphere

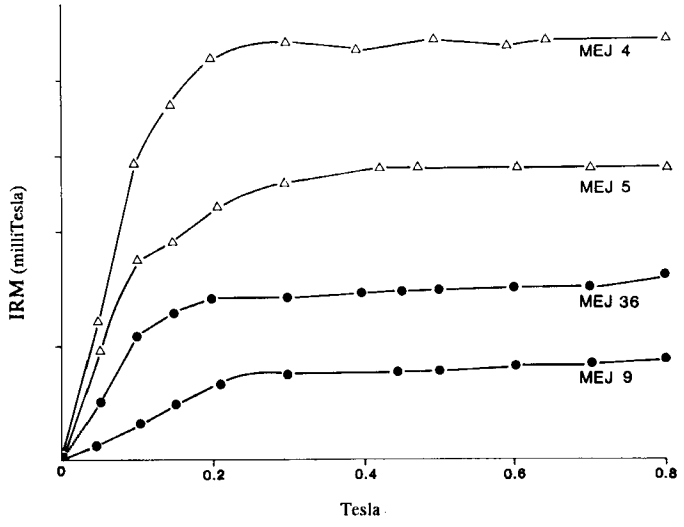


Figure 9. Representative IRM curves for samples from the Moreno Complex (circles) and Jorgina Formation (triangles). See text for explanation

Palaeozoic sedimentary, igneous and metamorphic rocks occur as isolated (usually fault-bounded) fragments throughout much of the north Chilean forearc (see Herve *et al.* 1981; Breitkreuz 1986; Damm *et al.* 1986; Baeza and Pichowiak 1988), isolated within the volcanic and plutonic rocks of the Jurassic magmatic province. This may indicate that the Andean orogenic cycle developed on fragmented Palaeozoic continental crust. The older strata of the Mejillones Peninsula could therefore represent a para-autochthonous fragment of continental crust rifted from the South American craton during a Permo-Triassic

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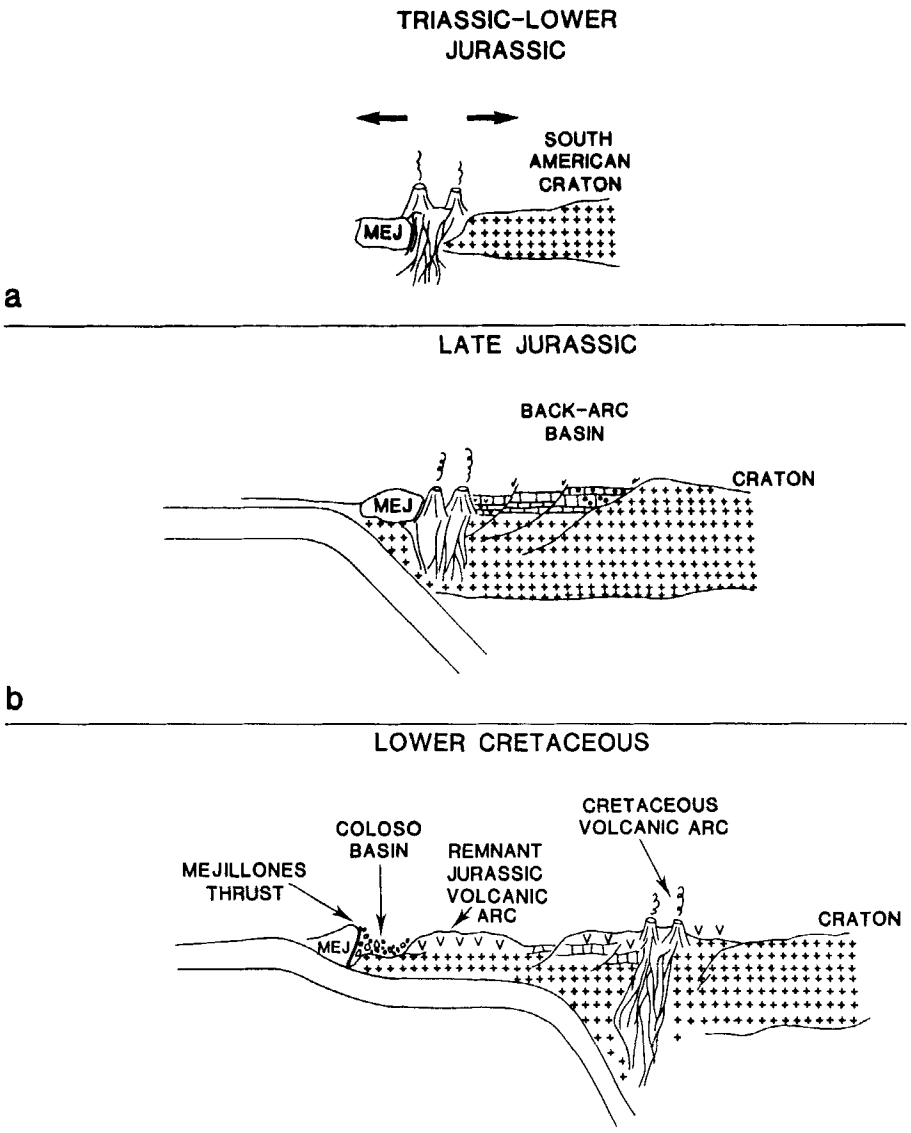


Figure 10. Proposed model for the evolution of the Mejillones Peninsula. (a) Origination as a piece of the South American craton rifted off during the Late Triassic (see Coira *et al.* 1982). The peninsula was emplaced by the Mid-Jurassic as shown in (b), where it was subject to metamorphism associated with intrusion of the north Chilean batholith. The carbonate-dominated backarc basin, which was developed in the Lower Jurassic following Triassic rifting (Coira *et al.* 1982; Hartley 1987), was inverted during the mid-Cretaceous Peruvian Orogeny. (c) Development of the Coloso Basin between the Jurassic volcanic province and the Mejillones Peninsula

extensional tectonic regime (Coira *et al.* 1982) and caught up within the ensialic Jurassic backarc basin (Rogers 1985) (Figure 10). Following emplacement, the Lower Palaeozoic strata were subjected to high grade contact metamorphism associated with the intrusion of the mid-Jurassic north Chilean coastal batholith, resulting in thermal remagnetization and resetting of radiometric dates. The suture zone between the Mejillones Peninsula and the volcanic are (represented by the Mejillones Thrust) was utilized in Early Cretaceous times as a sinistral shear zone (Scheuber and Andriessen 1990) during development of the Coloso Basin (Figure 10). The basin was infilled by proximal alluvial fan detritus derived from the Moreno Complex to the west (Flint *et al.* 1986; Hartley 1987; Flint and Turner 1988). Following these events, areas of the Cordillera de la Costa in northern Chile were subjected to significant clockwise rotation

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[see Beck (1988) for review] whereas localized, non-uniform rotation affected the Mejillones Peninsula. The contrasting styles of rotation probably reflect the different responses of the Cordillera de la Costa and the Mejillones Peninsula to the same rotation-generating tectonic event. The amount of sinistral movement on the Atacama Fault zone during the Late Mesozoic (Scheuber and Andriessen 1990) was not sufficient to result in the large-scale latitudinal displacement of crustal fragments (Hartley *et al.* 1988).

Irwin *et al.* (1987) noted a similar history for a high grade metamorphic complex from the Central Chilean Pacific margin between 32°15′ and 33°30′S. Their palaeomagnetic data suggested that the complex was an integral part of the Chilean margin by mid-Jurassic times, as 170 Ma dykes which cross-cut the complex do not indicate any rotation or latitudinal translation. Although it is not possible to exclude pre-Jurassic accretion of allochthonous terranes as an important orogenic process along the central Andean margin, it appears that no terrane accretion took place in northern Chile and possibly along the whole Central Andean Pacific margin, after the Jurassic.

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